

Storm Water Tracker

Recording every storm overflow discharge in England, and flagging the ones that started on a dry day.

1 · EXECUTIVE SUMMARY

The problem. Storm overflows are allowed to discharge sewage into rivers and the sea when heavy rain would otherwise overwhelm a combined sewer, one carrying both rainwater and sewage, and flood homes and streets. They are not supposed to discharge on dry days. A discharge with no storm to relieve suggests a fault, a blockage or too little capacity.

The tracker. Storm Water Tracker is a website that records every storm overflow discharge published by England's ten water companies, checks each one against Environment Agency rainfall data, and flags the ones that started on a dry day. It reads the companies' live feeds every ten minutes and keeps every discharge permanently.

The one rule it applies is the Environment Agency's own definition, published on 28 August 2024:

"A dry day spill is when a storm overflow is used on a 'dry day' – which is defined as no rainfall above 0.25mm on that day and the preceding 24 hours."

The site implements that sentence and nothing else. A discharge is flagged when the rain recorded near the overflow, over the day it started and the day before, totals 0.25 mm or less.

Key numbers · figures as at 24 September 2026

14,200

storm overflows tracked, all ten companies

93,326

discharges classified, 90,238 from Thames Water's history

12,025

dry day spills flagged, 693 of them contested by radar

98.6%

of flags rest on complete rain data (192 of 192 readings)

400

flagged (59 contested) of 3,088 discharges since 17 Sep 2026

69.2%

of the flags radar can check, it agrees with (3,785 of 5,472)

Most classified discharges are Thames Water's history back to April 2022, so the larger totals describe one company's area, not the country. The record for all ten companies began on 17 September 2026.

What makes it different.

- It applies the Environment Agency's own published definition exactly, not a rule of its own devising.
- It keeps a permanent, per-event record going forward; the official live feeds do not.
- It shows the evidence behind every flag and publishes the full data, so anyone can reproduce a flag from the Environment Agency's public data.
- It counts conservatively: the stricter reading of the rule, with radar-contested flags counted separately.

The limits. A flag means the rain measured near an overflow was at or below the Environment Agency's threshold; it does not mean an offence occurred. The Environment Agency lists exceptions and treats dry day spills as a potential breach until confirmed by investigation. The method misses discharges made while a monitor is offline, and a gauge up to 10 km away can miss a local shower. Company figures are not a ranking of performance, and the record is too young for trends or annual totals. Section 8 gives the limits in full.

A personal, non-commercial project. Not affiliated with any water company, regulator or campaign group. Flagged discharges are potential breaches of the Environment Agency's dry day definition; only the Environment Agency can confirm a breach.

2 Background: why this record exists

A combined sewer carries both rainwater and sewage. In heavy rain it can fill faster than the treatment works can take the flow. A storm overflow is a designed relief point: when the sewer is overwhelmed, the excess spills to a river or the sea rather than backing up into homes and streets. That is its purpose, and a discharge during a storm is what the system was built to do.

The concern is overflows discharging when it has *not* been raining. With no storm to relieve, a discharge suggests something else is wrong: a fault, a blockage, or a sewer without enough capacity for ordinary flows. The Environment Agency puts it directly: "Storm overflows should not spill on dry days, but there are exceptions."

The live feeds: times, but no history

Under section 81 of the Environment Act 2021, a sewerage company whose area is wholly or mainly in England must publish each storm overflow discharge promptly:

"The information referred to in subsection (1)(a) to (c) must be published within an hour of the discharge beginning; and that referred to in subsection (1)(d) within an hour of it ending."

That creates a live public feed, gathered for all ten companies by the National Storm Overflow Hub. But the feed is a snapshot. It shows each overflow's most recent discharge; when a new discharge begins, the previous one drops out of view. The Hub's own FAQ states: "The Hub does not have access to data from before its date of publication."

The annual returns: history, but no times

Separately, the Environment Agency publishes annual Event Duration Monitoring returns. Those are kept, but they count spills per overflow per year and carry no per-event start and end times. The House of Commons Library notes the consequence: "there is at present no way to distinguish 'dry spills' (discharge of sewage at times of little to no rainfall, normally in breach of permit conditions) within this dataset."

So the times exist but are not kept, and the kept records have no times. Storm Water Tracker sits in that gap. It reads the live feeds every ten minutes, keeps every discharge permanently, and applies the dry-day test to each one as it arrives.

3 The rule, in full

The Environment Agency's definition, quoted exactly from its blog post of 28 August 2024:

"A dry day spill is when a storm overflow is used on a 'dry day' – which is defined as no rainfall above 0.25mm on that day and the preceding 24 hours."

How the site implements that sentence

- **What is judged:** the moment a discharge *starts*. The relevant day is the calendar day, in UTC, on which it started. A discharge that began in rain and continued into a dry day is not flagged.
- **The window:** "that day and the preceding 24 hours" is read as the 48 hours from 00:00 UTC on the day before the start to 00:00 UTC on the day after it.
- **The measurement:** every 15-minute rainfall total recorded in that window at the chosen rain gauge is added up. A complete window is 192 readings.

- **The verdict:** the discharge is flagged as a dry day spill if, and only if, the 48-hour total is 0.25 mm or less.

Why the total, and why that matters

The Environment Agency's sentence can be read two ways: no *single* reading above 0.25 mm, or no more than 0.25 mm *in all*. The site uses the total, which is the stricter test for raising a flag. A window with three readings of 0.2 mm has no single reading above the threshold, but totals 0.6 mm: it would be flagged under the other reading, and is not flagged under this one. Using the total therefore flags *fewer* discharges and keeps false flags to a minimum. Both numbers, the window total and the largest single reading, are published for every event, so anyone can recompute the other reading.

Which rain gauge

Rainfall comes from the Environment Agency's Hydrology API. For each overflow the site uses the nearest gauge within 10 km that returned at least 176 of the expected 192 readings for the window. The 10 km limit is the site's own assumption, because the Environment Agency publishes no distance, and it is stated as such.

Where no gauge lies within 10 km, the site follows the only published regulatory instruction on the point. It comes from Natural Resources Wales, the Welsh regulator, not the Environment Agency: "Where there is no nearby rain gauge, the three closest gauges can be triangulated." The site combines the three closest usable gauges within 20 km, weighted by inverse distance. The guidance names no distance and no weighting; the 20 km limit and the inverse-distance weighting are the site's own choices. If fewer than three usable gauges lie within 20 km, the discharge is recorded as *No gauge within 10 km* and cannot be classified under this method.

Broken gauges

A failed rain gauge usually fails silently. A blocked funnel reports 0.00 mm however hard it rains, which looks exactly like a dry day and would produce a false flag. The site therefore skips a gauge that has recorded nothing for days while independent evidence (other gauges nearby, or radar) says it rained, and moves to the next-nearest. Each flagged discharge states how many gauges were skipped.

Time zone

Everything is in UTC. This is a stated assumption, and it was verified rather than assumed: on a British Summer Time day, eight gauges were compared against the Environment Agency's separate real-time rainfall API, and they matched on 96 of 96 readings with no offset.

4 How a discharge becomes a verdict

1. **Collect.** Every ten minutes, the ten companies' feeds are read through the National Storm Overflow Hub. New discharges are written to a permanent record.
2. **Wait for the window.** A verdict cannot be reached until the 48-hour rainfall window has closed.
3. **Fetch rainfall.** The Environment Agency's 15-minute gauge totals for the window are pulled. The Environment Agency publishes some readings late, so the rainfall may arrive over the following days.
4. **Classify.** The rule in section 3 is applied. Until the data can no longer change, the verdict is marked *provisional* and re-checked every hour.
5. **Finalise.** A verdict becomes final when it can no longer change: when all 192 readings are present and 72 hours have passed since the window closed, or when 14 days have passed regardless.

6. **Publish.** The site is rebuilt, and every flagged discharge gets its own page carrying its evidence.

A verdict typically appears one to three days after the discharge started. The delay comes from the rainfall data, not the discharge data. Every change of verdict between “dry day spill” and “not a dry day” is recorded in a public log that anyone can download.

The verdicts a discharge can receive

Verdict	Meaning
Dry day spill	The 48-hour rainfall total at the chosen gauge (or the three triangulated gauges) was 0.25 mm or less. Shown on the site with the badge “Dry day spill · EA definition” and the sub-line “Potential breach; not confirmed.”
Not a dry day	The 48-hour total was above 0.25 mm.
Rain check pending	The rainfall window has not yet closed, or rain data may still arrive.
Insufficient rain data	The nearest gauges returned fewer than 176 of the 192 readings.
No gauge within 10 km	No usable gauge within 10 km and too few within 20 km to triangulate, or the overflow’s location was not published. Cannot be classified under this method.

5 The evidence behind every flag

No flag is published without its evidence. Each flagged discharge shows:

- the overflow’s name and ID, and the water company;
- the receiving watercourse;
- the start and end times of the discharge;
- the rain gauge used and its distance from the overflow (and, where nearer gauges were skipped as broken, how many);
- the rainfall totals for the 48-hour window;
- how many of the 192 readings were present;
- the rule version that produced the verdict, and the timestamp of the classification.

The complete dataset is downloadable as CSV.

A reader can check any single flag against the Environment Agency’s own public API and reproduce it. The gauge and the window are named and the readings counted, so the calculation can be repeated independently from the Environment Agency’s own data.

6 Checking the checker: rainfall radar

A rain gauge measures rain falling into one funnel, which may be up to 10 km from the overflow (20 km where three gauges are combined). Met Office rainfall radar estimates rain over every square kilometre, including the overflow’s own. The two are independent, and the Environment Agency says it uses both, gathering “rainfall data from local rain gauges as well as rainfall radar information”.

For each flagged discharge, the site adds up radar rainfall over the same 48-hour window for the overflow’s own one-kilometre square, and for the wettest of the nine squares centred on it (its own square and the eight around it). Both are published beside the gauge evidence.

Contested flags

Where the radar recorded materially more rain than the gauge did (more than 1 mm over the window in the wettest of those nine squares, four times the rule’s own threshold, against a gauge total of 0.25 mm or less) the flag is called **contested**. On the site, contested flags are counted separately from the headline figure rather than added into it; in this document, every total of flags states how many of them are contested. This is a counting decision, not a rule change:

- A contested flag is **still published**. It keeps its page and its evidence, and it stays in every data file. Nothing is hidden; it is counted in its own column.
- A flag the radar **cannot** check is **not** contested. The radar archive reaches back only about two years, and an older discharge must not be demoted for something the archive cannot speak to.
- A contested flag is **not a withdrawn one**. The Environment Agency’s definition is written around rainfall measurements, and the site implements that definition. A contest is a reason to look closer.

The 1 mm bar is deliberate. Across the archive the median disagreement is 0.79 mm, which means most disagreements are simply a radar estimate landing on the other side of the same line: noise, not evidence.

Radar does not decide verdicts

This was tested, not assumed. Every event then held was classified twice, once with the radar data present and once with it removed, and every verdict was identical. (Radar can serve as independent evidence that a silent gauge is broken, which is why the test was run rather than the claim simply made.)

Radar and the flags it can check · figures as at 24 September 2026

Measure	Count
Dry day flags the radar archive can independently check	5,472
Of those, flags the radar agrees with (wettest of the nine squares 0.25 mm or less)	3,785 (69.2%)
Flags contested by radar, all classified discharges	693
Flags contested by radar, live record since 17 September 2026	59
Radar archive held	669 days, from 21 November 2024

7 What the tracker has found

Snapshot taken 24 September 2026, 03:24 UTC. The site collects new discharges every ten minutes, so these figures move on from the moment they were taken. The figures below are one internally consistent set and should not be mixed with numbers read from the live site on another day.

7.1 Scale of the record

Measure	Value
Water companies covered	10 (all of England)
Storm overflows tracked	14,200
Total discharges classified	93,326
of which: since the site's own record began (17 September 2026)	3,088
of which: Thames Water's published history, back to April 2022	90,238
Date the permanent record began	17 September 2026
Rule version in force	dry-day-v3
Radar archive held	669 days, from 21 November 2024

7.2 Verdicts across everything classified (93,326 discharges)

Verdict	Count
Not a dry day	78,003
Dry day spill	12,025 (of which 693 contested by radar)
Insufficient rain data	2,537
Rain check pending	520
No gauge within 10 km	241

Most of these discharges are Thames Water's own published history, which no other company provides. These totals are weighted heavily toward one company's area and are not a national picture.

7.3 The live record only: all ten companies, since 17 September 2026 (3,088 discharges)

Verdict	Count
Not a dry day	1,958
Dry day spill	400 (of which 59 contested by radar)
Rain check pending	520
No gauge within 10 km	109
Insufficient rain data	101

This is the only part of the record in which every company's feed is collected in the same way. It spans days, not months: it supports no annual total and no trend.

7.4 Dry day spills by company, live record since 17 September 2026 (uncontested flags only)

Company	Dry day spills
South West Water	83
Severn Trent Water	79
Anglian Water	72
Northumbrian Water	33
United Utilities	21
Wessex Water	17
Thames Water	15
Yorkshire Water	14
Southern Water	7
ST Connect	0

These figures are not a ranking of companies.

Differences between companies reflect, in unknown proportion: how many overflows each company operates; where those overflows sit relative to working rain gauges; local geography and rainfall; differences in how companies detect and report discharges; and monitor reliability. They are not a measure of environmental performance and must not be read as one. ST Connect currently publishes only a placeholder feed on the Hub, so its zero reflects the absence of published data, not the absence of discharges.

7.5 Dry day spills by region of England, live record since 17 September 2026 (uncontested flags only)

Region	Dry day spills	Overflows behind them
South West	117	30
West Midlands	51	7
East Midlands	47	26
South East	33	10
North East	32	19
East of England	26	20
Yorkshire and The Humber	15	10
North West	12	10
London	8	3
All nine regions	341	135

These figures are not a ranking of regions.

Differences between regions reflect, in unknown proportion: how many overflows each region contains; where those overflows sit relative to working rain gauges; local geography and rainfall; which companies serve the region and how they detect and report discharges; and monitor reliability. A few overflows flagged repeatedly can dominate a region's count, which is why the right-hand column shows how many distinct overflows lie behind each figure. They are not a measure of environmental performance and must not be read as one.

How discharges were placed in regions. The regions are England's nine official ONS regions (December 2022), as drawn on the site's map. The site does not assign overflows to regions itself: these figures were produced for this document from the same snapshot, and they sum to the total of table 7.4. Each overflow's published coordinates were tested against the ONS full-resolution boundaries: 306 flags fell inside the boundaries clipped to the coastline, 31 at coastal and estuary outfalls inside the "extent of the realm" boundaries, and 4, at three outfalls just offshore of both, were placed in the only region within about 2 km of them (none other lies within 10 km). The simplified boundary file used to draw the map was not used: it is too coarse near regional borders.

7.6 Quality of the evidence

Measure	Value
Dry day flags resting on a complete 48-hour rainfall window (192 of 192 readings)	98.6%
Dry day flags the radar archive can independently check	5,472
Of those, flags the radar agrees with	3,785 (69.2%)

What these figures do not show

- No national annual total of dry day spills: the record began on 17 September 2026.
- No comparison of companies' or regions' performance (see the notes under 7.4 and 7.5).
- No trend, increase or decrease over time (the live record is days old), and no rate per overflow, company or region.

8 What this method can miss

8.1 Discharges the tracker may never see, or see imperfectly

- The feeds show only each overflow's **most recent** discharge. If a discharge starts and ends between two ten-minute reads, and another begins before the next read, the first is never seen.
- Where a discharge's end is never observed, it is inferred from the start of the next one and labelled as inferred.
- Where a company publishes an end time earlier than its start time, the end is left blank rather than guessed.
- Monitors go offline. An overflow discharging while its monitor is offline never appears at all.
- The feed data are the companies' own, unaudited. The Hub states: "Data provided on the map and in the API is near real-time data and has not undergone an audit process required to comply with the Environment Agency regulatory EDM Annual Return dataset."

8.2 Rainfall the tracker may misjudge

- A gauge up to 10 km away (up to 20 km where three gauges are combined) can miss a local shower that fell on the overflow, or record one that missed it.
- Overflows with no usable gauge within range cannot be classified under this method.
- A "day" is a UTC calendar day. This is a documented assumption, verified against the Environment Agency's real-time rainfall data, but it is still an assumption.

8.3 What a flag is, and is not

A dry day spill flag means: **the rainfall measured near this overflow, over the 48 hours defined by the Environment Agency, was at or below the Environment Agency's own threshold.** It does not mean an offence occurred. The Environment Agency lists exceptions, and says it will "treat them as a potential breach until we have confirmed through further investigation". Only the regulator can confirm a breach.

8.4 Why company and regional figures must not be read as a ranking

Differences between companies in table 7.4 reflect, in unknown proportion: how many overflows each company operates, where those overflows sit relative to working rain gauges, the local geography and rainfall, differences in how companies detect and report discharges, and monitor reliability. The same holds for the regional figures in table 7.5, where a few overflows flagged repeatedly can also dominate a region's count. They are **not** a ranking of environmental performance.

8.5 The record is young

The site's own permanent record began on **17 September 2026**. Most of the classified volume in table 7.2 is Thames Water's own published history, which no other company provides. The totals in table 7.2 are therefore weighted heavily toward one company's geography and must never be described as a national picture. For a statement covering all ten companies, table 7.3 is the right source, and it covers a period of days.

9 Sources

Each source was fetched on the date shown. Every link and quotation below was re-checked against the live source on 24 September 2026. Quotations are verbatim.

1. Environment Agency blog, "What are dry day spills?", 28 August 2024

Fetches 14–15 September 2026

<https://environmentagency.blog.gov.uk/2024/08/28/what-are-dry-day-spills>

"A dry day spill is when a storm overflow is used on a 'dry day' – which is defined as no rainfall above 0.25mm on that day and the preceding 24 hours."

"Storm overflows should not spill on dry days, but there are exceptions."

"rainfall data from local rain gauges as well as rainfall radar information"

"treat them as a potential breach until we have confirmed through further investigation"

2. Environment Agency, storm overflow spill data for 2024, 27 March 2025

Fetches 15 September 2026

<https://www.gov.uk/government/news/environment-agency-storm-overflow-spill-data-for-2024>

"since January, all day dry spills – no matter how small – are now classified as pollution incidents"

3. Environment Agency, 2025 monitoring data release, 26 March 2026

Fetches 15 September 2026

[https://www.gov.uk/government/news/](https://www.gov.uk/government/news/fewer-and-shorter-storm-overflow-spills-in-2025-new-monitoring-data-shows)

[fewer-and-shorter-storm-overflow-spills-in-2025-new-monitoring-data-shows](https://www.gov.uk/government/news/fewer-and-shorter-storm-overflow-spills-in-2025-new-monitoring-data-shows)

"There were 291,492 spill events in 2025, a 35% reduction on 2024."

"Every single storm overflow in England now has an event duration monitor fitted, providing the most complete national picture to date."

4. Environment Act 2021, section 81

Fetches 15 September 2026

<https://www.legislation.gov.uk/ukpga/2021/30/section/81>

"The information referred to in subsection (1)(a) to (c) must be published within an hour of the discharge beginning; and that referred to in subsection (1)(d) within an hour of it ending."

5. National Storm Overflow Hub (Stream), storm overflows data page

Fetches 15 September 2026

<https://www.streamwaterdata.co.uk/pages/storm-overflows-data>

"Data provided on the map and in the API is near real-time data and has not undergone an audit process required to comply with the Environment Agency regulatory EDM Annual Return dataset."

6. National Storm Overflow Hub (Stream), FAQ

Fetches 15 September 2026

<https://www.streamwaterdata.co.uk/pages/storm-overflows-faqs>

"The Hub does not have access to data from before its date of publication."

7. House of Commons Library, CBP-10027, "Sewage discharges", 31 March 2026

Fetches 15 September 2026

<https://researchbriefings.files.parliament.uk/documents/CBP-10027/CBP-10027.pdf>

"there is at present no way to distinguish 'dry spills' (discharge of sewage at times of little to no rainfall, normally in breach of permit conditions) within this dataset."

8. Natural Resources Wales, guidance note GN066, "How to classify storm overflow performance", 26 October 2023

Fetches 20 September 2026

<https://afonyddcymru.org/wp-content/uploads/2024/03/GN066-How-to-classify-storm-overflow-performance.pdf>

"Use rain gauge data that is the most representative for the SO. Where there is no nearby rain gauge, the three closest gauges can be triangulated."

Natural Resources Wales is the Welsh regulator, not the Environment Agency. Its guidance is cited because it is the closest published prescription either regulator has issued on the point.

9. Environment Agency Hydrology API

Fetches 15 September 2026

<https://environment.data.gov.uk/hydrology/doc/reference>

Open Government Licence v3.0. The source of the 15-minute rainfall totals behind every verdict.

10. Met Office UK radar observations, Registry of Open Data on AWS

Fetches 15 September 2026

<https://registry.opendata.aws/met-office-uk-radar-observations/>

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11. Office for National Statistics, Regions (December 2022) Boundaries EN, full-resolution versions

Queried 24 September 2026, clipped to the coastline (BFC) and to the extent of the realm (BFE)

[https://services1.arcgis.com/ESMARspQHMYMw9BZ9/arcgis/rest/services/](https://services1.arcgis.com/ESMARspQHMYMw9BZ9/arcgis/rest/services/Regions_December_2022_Boundaries_EN_BFC_V2/FeatureServer)

[Regions_December_2022_Boundaries_EN_BFC_V2/FeatureServer](https://services1.arcgis.com/ESMARspQHMYMw9BZ9/arcgis/rest/services/Regions_December_2022_Boundaries_EN_BFC_V2/FeatureServer)

[https://services1.arcgis.com/ESMARspQHMYMw9BZ9/arcgis/rest/services/](https://services1.arcgis.com/ESMARspQHMYMw9BZ9/arcgis/rest/services/Regions_December_2022_EN_BFE/FeatureServer)

[Regions_December_2022_EN_BFE/FeatureServer](https://services1.arcgis.com/ESMARspQHMYMw9BZ9/arcgis/rest/services/Regions_December_2022_EN_BFE/FeatureServer)

Used only to place flagged discharges in regions for table 7.5. Open Government Licence v3.0.

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Rainfall: Environment Agency Hydrology API, Open Government Licence v3.0.

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Region boundaries: Source: Office for National Statistics licensed under the Open Government Licence v.3.0. Contains OS data © Crown copyright and database right 2022.